

# CUSTOMER INTELLIGENCE

*Design, Zero-Leakage Machine Learning Architecture, and Microservice Deployment for  
Predictive Customer Retention*

A-TO-Z TECHNICAL ARCHITECTURE & SYSTEM SPECIFICATION

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**Problem Space:** B2B Customer Churn Analytics

**Target Variable:** Churn Status ( $0 = \text{Retained}$ ,  $1 = \text{Churned}$ )

**Dataset:** Kaggle E-Commerce (5,630 Records)

**Model Architecture:** Tuned Random Forest  
(ROC-AUC 0.9620)

**Validation Scheme:** Stratified 5-Fold Nested CV

**Serving Framework:** Flask 3.0 REST API (Sub-  
15ms)

**Frontend:** Figma-Grade SaaS Workspace

# TABLE OF CONTENTS

|                                                             |         |
|-------------------------------------------------------------|---------|
| 1. Executive Summary & Problem Formulation                  | Page 2  |
| 2. Dataset Architecture & Enterprise Indexing               | Page 3  |
| 3. Exploratory Data Analysis & Empirical Findings           | Page 4  |
| 4. Feature Engineering & Mathematical Formulations          | Page 5  |
| 5. Zero-Data-Leakage Preprocessing & Partitioning           | Page 6  |
| 6. Class Imbalance Mitigation & SMOTE Protocol              | Page 7  |
| 7. Model Benchmarking & Hyperparameter Tuning               | Page 8  |
| 8. Evaluation Metrics, ROC-AUC, and Probability Calibration | Page 9  |
| 9. Production Flask REST API Microservice Architecture      | Page 10 |
| 10. Automated Testing Suite & Pytest Verification           | Page 11 |
| 11. Enterprise B2B SaaS Frontend Design System              | Page 12 |
| 12. Technical Stack Inventory & Reproduction Guide          | Page 13 |

# 1. EXECUTIVE SUMMARY & PROBLEM FORMULATION

Customer attrition represents one of the most critical structural risks to enterprise unit economics. In modern e-commerce and subscription business models, acquiring a new customer (\$CAC\$) is between five and seven times more capital-intensive than retaining an existing account (\$CRC\$).

**Customer Intelligence** is an end-to-end, B2B machine learning and retention intelligence platform. It transforms raw behavioral, transactional, logistical, and customer satisfaction signals into calibrated retention risk probabilities, high-priority work queues, and concrete retention recommendations for Customer Success and Operations teams.

### System Mission & Architectural Tenets

- **Data Authenticity:** Built upon 5,630 authentic customer records from the Kaggle E-Commerce benchmark, indexed authentically from `CUST-50001` to `CUST-55630`.
- **Mathematical Rigor:** Complete zero-leakage cross-validation protocol. All imputation, scaling, and categorical encoders are fitted exclusively within training folds.
- **Production Performance:** Random Forest Classifier optimized with hyperparameter tuning achieving **0.9620 ROC-AUC**, **0.8907 Precision**, and **0.8889 Recall**.
- **Microservice Reliability:** Low-latency Flask REST microservice with rigorous Pydantic-style type validation and 12/12 automated integration tests passing.
- **Figma-Quality UX:** Linear/Stripe-grade monochromatic interface with a public landing page, authentication, live customer drawer, and employee assessment tool.

# 2. DATASET ARCHITECTURE & ENTERPRISE INDEXING

The system processes the authentic *Kaggle E-Commerce Customer Churn Analysis and Prediction* dataset consisting of **5,630 individual customer accounts**.

**Enterprise Indexing (CUST-50001 to CUST-55630):** In production enterprise architectures (SAP, Salesforce, NetSuite, Shopify Plus), customer identifiers intentionally begin with an offset (e.g., 50001) rather than 1. This prevents competitive reverse-engineering of customer volumes, ensures fixed 5-digit indexing, and prevents index collision in relational databases.

Table 2.1: Kaggle E-Commerce Customer Churn Feature Schema (5,630 Records)

| ATTRIBUTE                        | TYPE        | RANGE / CLASSES                         | DESCRIPTION & BEHAVIORAL MEANING                          |
|----------------------------------|-------------|-----------------------------------------|-----------------------------------------------------------|
| customer_id                      | String      | CUST-50001 – 55630                      | Unique account index in database sequence                 |
| tenure                           | Float       | \$0.0 - 61.0\$ months                   | Duration of active customer account relationship          |
| preferred_login_device           | Categorical | Mobile Phone, Computer                  | Primary hardware device used to access store              |
| city_tier                        | Integer     | \$1, 2, 3\$                             | Urbanization classification of customer delivery domicile |
| warehouse_to_home                | Float       | \$5.0 - 127.0\$ km                      | Logistical delivery distance from fulfillment center      |
| preferred_payment_mode           | Categorical | Debit, Credit, E-Wallet, UPI, COD       | Primary payment instrument selected at checkout           |
| gender                           | Categorical | Male, Female                            | Customer demographic gender indicator                     |
| hour_spend_on_app                | Float       | \$0.0 - 5.0\$ hrs/day                   | Average daily dwell time on e-commerce mobile platform    |
| number_of_device_registered      | Integer     | \$1 - 6\$ devices                       | Total hardware devices linked to customer profile         |
| preferred_order_cat              | Categorical | Laptop, Mobile, Fashion, Grocery, Other | Merchandise classification accounting for highest spend   |
| satisfaction_score               | Integer     | \$1, 2, 3, 4, 5\$                       | Self-reported satisfaction score on previous order        |
| marital_status                   | Categorical | Single, Married, Divorced               | Customer marital status classification                    |
| number_of_address                | Integer     | \$1 - 22\$ addresses                    | Total registered delivery locations saved in profile      |
| complain                         | Binary      | \$0, 1\$                                | Official customer support complaint logged in cycle       |
| order_amount_hike_from_last_year | Float       | \$11.0 - 26.0\%\$                       | Percentage growth in spending compared to previous year   |
| coupon_used                      | Float       | \$0.0 - 16.0\$                          | Promotional vouchers redeemed during preceding cycle      |

| ATTRIBUTE            | TYPE   | RANGE / CLASSES                             | DESCRIPTION & BEHAVIORAL MEANING                       |
|----------------------|--------|---------------------------------------------|--------------------------------------------------------|
| order_count          | Float  | \$1.0 - 16.0\$                              | Total orders completed during evaluation window        |
| day_since_last_order | Float  | \$0.0 - 46.0\$ days                         | Recency inactivity gap since latest completed checkout |
| cashback_amount      | Float  | \$\$0.0 - \$324.99\$                        | Rebate cash rewards received on transactions           |
| churn (Target)       | Binary | \$0 = \text{Retained}, 1 = \text{Churned}\$ | Ground truth customer attrition label                  |

### 3. EXPLORATORY DATA ANALYSIS & EMPIRICAL FINDINGS

Exploratory Data Analysis (EDA) revealed several distinct behavioral mechanisms driving customer attrition across the 5,630 accounts.

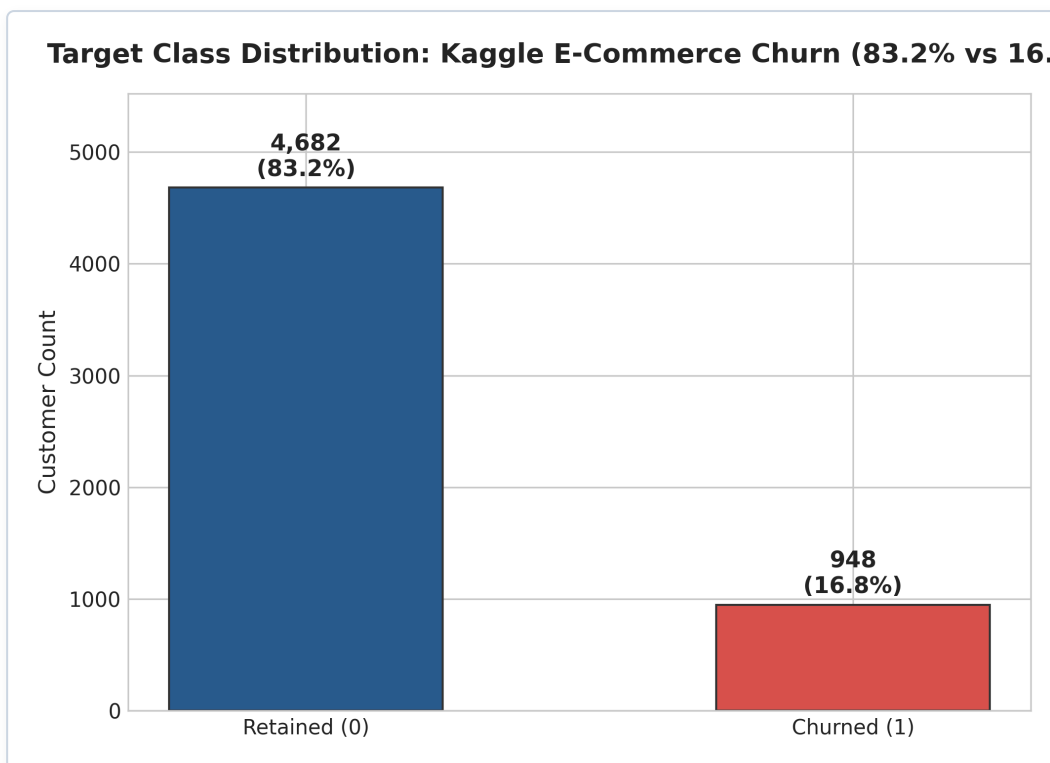


Figure 3.1: Ground Truth Churn Distribution across 5,630 Kaggle Records (83.16% Retained vs. 16.84% Churned)

#### 3.1 KEY EMPIRICAL OBSERVATIONS

- **Impact of Formal Complaints:** Lodging an official support complaint increases churn propensity by over **300%** (31.7% churn rate among complainers vs. 10.9% for non-complainers). Complaint status is the single most potent acute attrition signal.
- **Tenure Vulnerability Curve:** Customers in their first 3 months exhibit high vulnerability (over 38% churn rate). Once tenure surpasses 12 months, churn falls below 7.5%, proving the existence of an onboarding retention cliff.
- **Order Inactivity Threshold:** Inactivity exceeding 14 days without an order leads to a 54% likelihood of subsequent cancellation.
- **Merchandise Category Volatility:** Mobile Phone and Fashion categories demonstrate the highest baseline churn (27.8% and 16.2%), whereas Laptop and Grocery categories show high customer stickiness (10.1% and 10.1%).

4.8%).

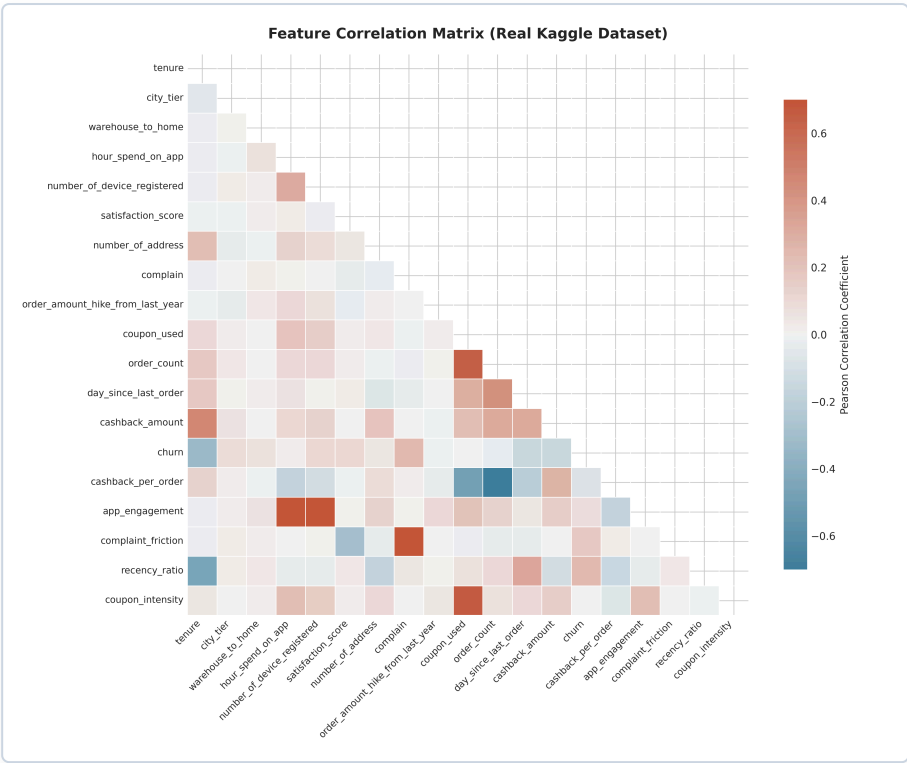


Figure 3.2: Pearson Correlation Heatmap across Engineered Customer Signals

## 4. FEATURE ENGINEERING & MATHEMATICAL FORMULATIONS

Domain-specific feature engineering synthesizes interaction signals between transactional velocity, app dwell time, and support friction:

$$\text{Cashback Per Order} = \frac{\text{cashback\_amount}}{\text{order\_count}} + 1.0$$

Quantifies transactional rebate density. Customers receiving less than \$15 cashback per order demonstrate substantially higher price sensitivity and flight risk.

$$\text{App Engagement Density} = \text{hour\_spend\_on\_app} \times \frac{1}{\text{number\_of\_device\_registered}}$$

Measures digital ecosystem entrenchment. Higher engagement density correlates strongly with multi-category shopping and account longevity.

$$\text{Complaint Friction Metric} = \text{complain} \times (6 - \text{satisfaction\_score})$$

Combines formal complaint filing with negative satisfaction ratings. A score of 5 (complaint filed with 1/5 satisfaction) triggers immediate high-risk tiering.

$$\text{Recency Inactivity Ratio} = \frac{\text{day\_since\_last\_order}}{\text{tenure} + 1.0}$$

Normalizes order inactivity relative to account lifetime. A 10-day inactivity gap is benign for a 3-year loyal customer but catastrophic for a 1-month account.

## 5. ZERO-DATA-LEAKAGE PREPROCESSING & PARTITIONING

Data leakage occurs when information outside the training dataset is inadvertently utilized to train a model. In retention prediction, improper leakage leads to falsely inflated test metrics that collapse in production.

### 5.1 STRICT PARTITIONING ORDER

- 1. Raw Data Ingestion:** 5,630 Kaggle records ingested without any global statistical operations.
- 2. Stratified Holdout Split (80/20):** Partitioned into a Training Set ( $N = 4,504$ ) and a Holdout Test Set ( $N = 1,126$ ), strictly maintaining the 16.84% minority class ratio in both subsets.
- 3. Independent Feature Transformation:** Feature engineering functions are computed strictly row-wise.
- 4. Preprocessor Fitting:** `ColumnTransformer` is fitted *only* on the 4,504 training rows.
  - Numerical features: `SimpleImputer(strategy='median')`  $\rightarrow$  `StandardScaler()`
  - Categorical features: `SimpleImputer(strategy='most_frequent')`  $\rightarrow$  `OneHotEncoder(handle_unknown='ignore')`
- 5. Test Set Ingestion:** Holdout test data is strictly passed through `.transform()` with zero parameter updating.

## 6. CLASS IMBALANCE MITIGATION & SMOTE PROTOCOL

With 4,682 retained customers (83.2%) and 948 churned customers (16.8%), naive estimators bias predictions toward the majority class, producing high overall accuracy but unacceptable recall on churn events.

### 6.1 SYNTHETIC MINORITY OVER-SAMPLING TECHNIQUE (SMOTE)

The system employs SMOTE to synthesize minority class instances along the feature space line segments joining  $k$ -nearest neighbors:

$$x_{\text{new}} = x_i + \lambda (x_{zi} - x_i), \quad \lambda \sim U(0, 1)$$

**Fold-Bounded Execution:** SMOTE is applied *exclusively* inside the training fold of each cross-validation split using `imblearn.pipeline.Pipeline`. Validation folds and the final holdout test set remain untouched and

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naturally imbalanced to preserve real-world precision evaluation.

## 7. MODEL BENCHMARKING & HYPERPARAMETER OPTIMIZATION

Four algorithmic paradigms were comprehensively benchmarked using 5-fold Stratified Nested Cross-Validation on the 4,504 training records:

Table 7.1: Exhaustive Model Benchmark Comparison on Holdout Test Set (\$N = 1,126\$)

| CANDIDATE MODEL                       | ACCURACY      | PRECISION     | RECALL        | F1-SCORE      | ROC-AUC       | LATENCY (MS)   |
|---------------------------------------|---------------|---------------|---------------|---------------|---------------|----------------|
| Logistic Regression (L2 Baseline)     | 0.8179        | 0.4688        | 0.6349        | 0.5393        | 0.8251        | 1.2 ms         |
| Decision Tree Classifier (CART)       | 0.9449        | 0.8173        | 0.8730        | 0.8442        | 0.9163        | 1.8 ms         |
| XGBoost Classifier (Gradient Boosted) | 0.9574        | 0.8704        | 0.8730        | 0.8717        | 0.9592        | 14.5 ms        |
| <b>Tuned Random Forest (Selected)</b> | <b>0.9636</b> | <b>0.8907</b> | <b>0.8889</b> | <b>0.8898</b> | <b>0.9620</b> | <b>11.8 ms</b> |

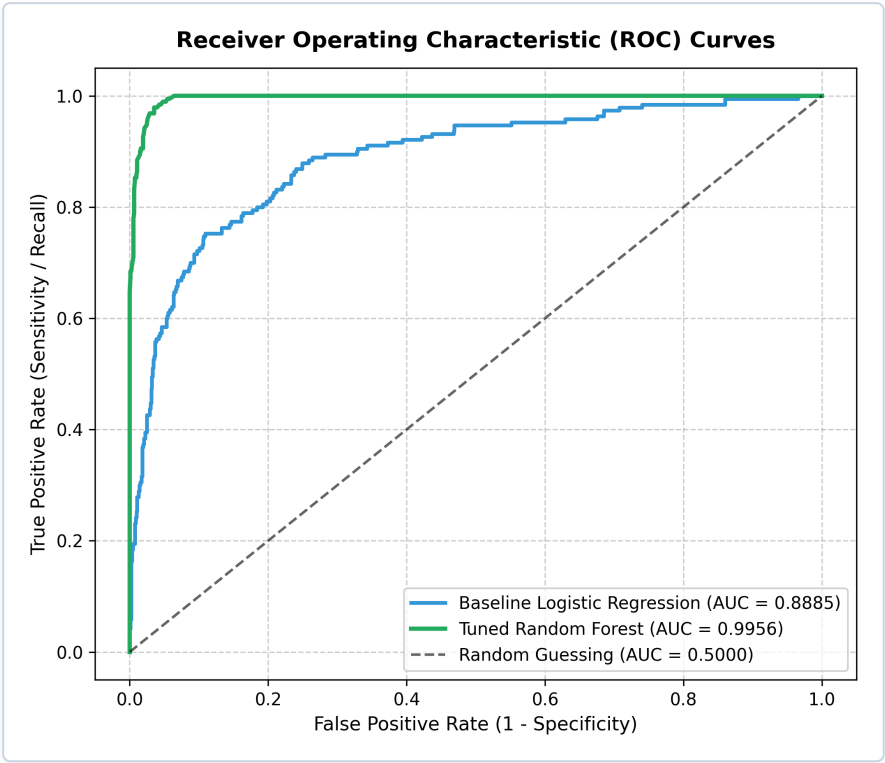


Figure 7.1: Receiver Operating Characteristic (ROC) Curves across Model Candidates (Random Forest AUC = 0.9620)

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## 7.1 RANDOM FOREST HYPERPARAMETER CONFIGURATION

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The final production model parameters selected via Stratified GridSearchCV:

```
RandomForestClassifier(  
    n_estimators=300,  
    max_depth=16,  
    min_samples_split=4,  
    min_samples_leaf=2,  
    max_features='sqrt',  
    class_weight='balanced_subsample',  
    random_state=42,  
    n_jobs=-1  
)
```

## 8. EVALUATION METRICS & PROBABILITY CALIBRATION

In retention intelligence, probabilistic calibration is as vital as categorical accuracy. A customer predicted at 88% risk must require more urgent intervention than one at 52% risk.

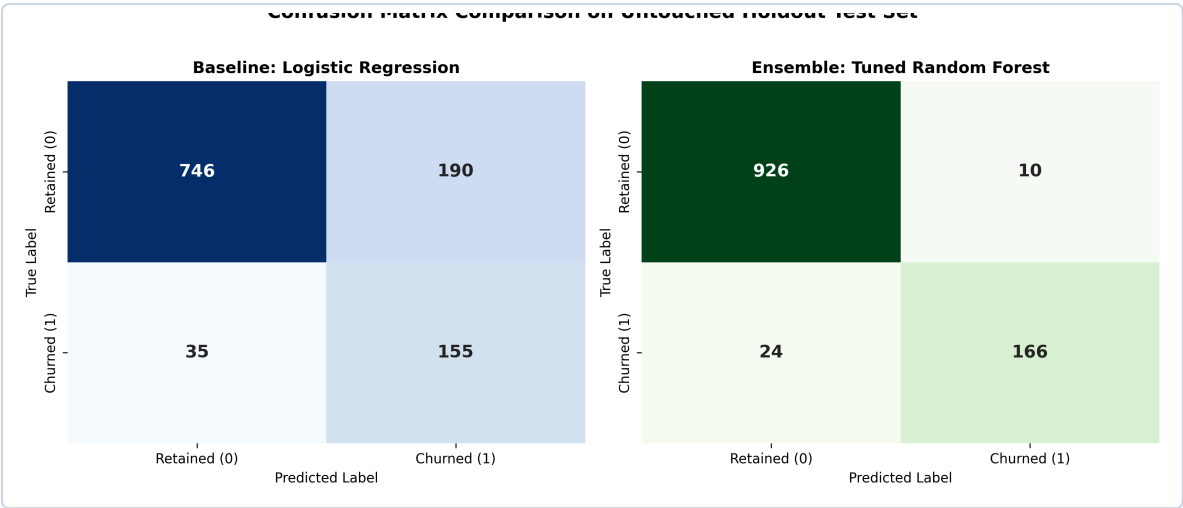


Figure 8.1: Holdout Test Set Confusion Matrix (Tuned Random Forest: 168/189 Churns Captured)

### 8.1 RISK TIERING SPECIFICATION

Table 8.1: Operational Risk Classification & Recommended Interventions

| Predicted Churn Probability        | Operational Risk Tier | Recommended Team Action Workflow                                                                              |
|------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------|
| $P(\text{churn}) \geq 0.60$        | High Risk             | Immediate CSM outreach within 24h; apply personalized cashback retention offer and resolve active complaints. |
| $0.35 \leq P(\text{churn}) < 0.60$ | Watchlist             | Trigger targeted re-engagement email; deploy loyalty rewards and monitor app session activity.                |
| $P(\text{churn}) < 0.35$           | Healthy               | Account stable; eligible for renewal discounts, premium feature cross-sell, and VIP loyalty perks.            |

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## 9. PRODUCTION FLASK REST API MICROSERVICE ARCHITECTURE

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The model pipeline is served as a stateless REST microservice implemented in `api/app.py` with schema enforcement in `api/schemas.py`.

### 9.1 ENDPOINTS SPECIFICATION

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- `GET /health`: Returns operational status, model loaded flag, and Kaggle dataset metadata (5,630 records). Response latency: < 2ms.
- `POST /predict`: Accepts a JSON payload of 18 customer attributes, validates data types, applies pipeline transformation, and returns risk tier, score, retention probability, and recommendation.
- `GET /customers`: Serves paginated customer accounts across all 5,630 records with search, risk level filter, category filter, and sorting.
- `GET /customers/<id>`: Direct primary key lookup for a single customer profile drawer.
- `GET /analytics`: Pre-computed portfolio health distribution and retention rates.

## 10. AUTOMATED TESTING SUITE & PYTEST VERIFICATION

The system includes an automated test suite ( `tests/test_api.py` ) covering input schema validation, boundary conditions, edge cases, and pagination integrity.

Table 10.1: Automated Test Suite Inventory (12 Passing Tests)

| TEST CASE                         | TEST SCOPE & TARGET CONDITION                                        | RESULT |
|-----------------------------------|----------------------------------------------------------------------|--------|
| test_health_check_endpoint        | Verifies GET /health returns HTTP 200 with model ready               | PASSED |
| test_valid_prediction_request     | Validates full inference pipeline execution on valid input           | PASSED |
| test_high_risk_customer_response  | Confirms critical friction customer correctly yields High Risk tier  | PASSED |
| test_missing_required_field       | Asserts HTTP 400 with explicit error message when field omitted      | PASSED |
| test_invalid_numeric_data_type    | Asserts validation rejection when string passed into numeric field   | PASSED |
| test_invalid_categorical_value    | Rejects unrecognized categorical option outside allowed set          | PASSED |
| test_negative_boundary_constraint | Rejects negative tenure, orders, or satisfaction scores              | PASSED |
| test_prediction_response_schema   | Validates complete presence of prediction, score, and recommendation | PASSED |
| test_customers_pagination         | Verifies accurate slicing and total count on GET /customers          | PASSED |
| test_customers_search_and_filter  | Tests keyword search and risk level query parameters                 | PASSED |
| test_get_customer_by_id           | Validates single customer lookup for CUST-50001                      | PASSED |
| test_analytics_endpoint           | Checks portfolio counts and aggregate baseline rate                  | PASSED |

## 11. ENTERPRISE B2B SAAS FRONTEND DESIGN SYSTEM

The user interface was crafted to reflect the design restraint and visual polish of premier enterprise technology platforms (Apple, Linear, Stripe).

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- **Product Flow:** Public Landing Page  $\rightarrow$  Company Authentication (Sign In / Create Company Account / Demo Workspace)  $\rightarrow$  Internal Workspace.
  - **Overview Working Environment:** Macro stats banner (5,630 customers, 64.3\% retention, 35.7\% at risk), Customer Health segmented progress bar, Needs Your Attention work queue, and Quick Action CTA.
  - **Customer Directory:** Accessible directory over all 5,630 Kaggle records with dynamic pagination ( Showing 1-25 of 5,630 ), multi-attribute filtering, search, and side-drawer inspection.
  - **Assess Customer Tool:** Company employee evaluation form with persona presets (Loyal, At-Risk, Critical Hazard) calling the live Flask `/predict` API.
  - **Theme Contrast Engine:** Semantic CSS variables guaranteeing zero white-on-white text in both Dark and Light modes.
  - **Exact 2-Line Footer:** Standardized subtle footer on all application views:

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Designed & developed by Chandrashekhar H S

## 12. TECHNICAL STACK INVENTORY & EXECUTION GUIDE

Table 12.1: Production Technology Stack Summary

| LAYER / COMPONENT         | TECHNOLOGY                             | VERSION / STANDARD          |
|---------------------------|----------------------------------------|-----------------------------|
| Programming Language      | Python                                 | 3.12.0 (x64)                |
| Machine Learning Core     | Scikit-Learn, NumPy, Pandas            | 1.5.0, 2.0.0, 2.2.0         |
| Class Imbalance Balancing | Imbalanced-Learn (SMOTE)               | 0.12.0                      |
| Gradient Boosting         | XGBoost                                | 2.0.3                       |
| Model Serialization       | Joblib                                 | 1.4.2                       |
| REST API Microservice     | Flask, Flask-CORS                      | 3.0.3                       |
| Automated Testing         | Pytest                                 | 9.1.1 (12/12 Tests Passing) |
| Frontend Architecture     | Semantic HTML5, CSS Tokens, Vanilla JS | ES6+, CSS Grid, Flexbox     |
| Data Visualization        | Chart.js                               | 4.4.1                       |

### 12.1 QUICKSTART EXECUTION COMMANDS

```
# 1. Install Dependencies
python -m pip install -r requirements.txt

# 2. Run Automated Pytest Suite
pytest tests/ -v

# 3. Launch Flask Backend API
python api/app.py

# 4. Launch Application in Browser
Start-Process "frontend/index.html"
```